Agent Memory **Evaluation Metrics and Formulas**

Agent memory evaluation uses **many metrics, not one**: the literature consistently measures retrieval quality, answer correctness, update/forgetting behavior, downstream use, and system cost separately, because end-task success alone does not reveal whether memory was written, retrieved, ranked, updated, or used correctly (Huang et al., 2026; Zhou et al., 2026; Dong et al., 2026; Tao et al., 2026).

1. Introduction

Recent benchmarks converge on the view that agent memory is a multi-stage capability spanning writing, maintenance, retrieval, and use across long-horizon interactions (Huang et al., 2026; Zhou et al., 2026; Liu et al., 2026; Du, 2026). Older work emphasized question-answer accuracy or conversational recall, but newer work argues that this misses dynamic updates, interference, selective forgetting, privacy leakage, and operational overhead (Zhang et al., 2024; Uddin et al., 2026; Jiang et al., 2026; Huang et al., 2026). The strongest shared message is methodological: no single scalar score adequately captures memory quality (Huang et al., 2026; Alqithami, 2025; Dong et al., 2026; Alake et al., 2026).

The metrics themselves fall into a few recurring families. Most papers still report some combination of Accuracy, F1, Recall@K, MRR, nDCG, MAP, or task success (Huang et al., 2026; Cherif, 2026; Shen et al., 2026; Yu et al., 2026). But newer benchmarks add metrics for selective forgetting, conflict handling, evidence use, cross-session learning, privacy leakage, temporal stability, token footprint, latency, and write-path cost (Uddin et al., 2026; Tao et al., 2026; Liu et al., 2026; Chen et al., 2026; Liu et al., 2026).

Do agent memory evaluations require multiple metrics beyond accuracy alone?

Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

FIGURE 1 Consensus on multidimensional agent memory evaluation

2. Methods

This synthesis draws on the supplied Deep Search corpus of agent-memory benchmark, survey, and systems papers indexed in Consensus across Semantic Scholar, PubMed, and other scholarly sources.

Priority went to papers that explicitly define evaluation metrics or formulas, introduce new metric families, or critique common evaluation blind spots. When a paper named a metric without a formula, it is included as evidence for usage, not for mathematical specification.

Search Strategy

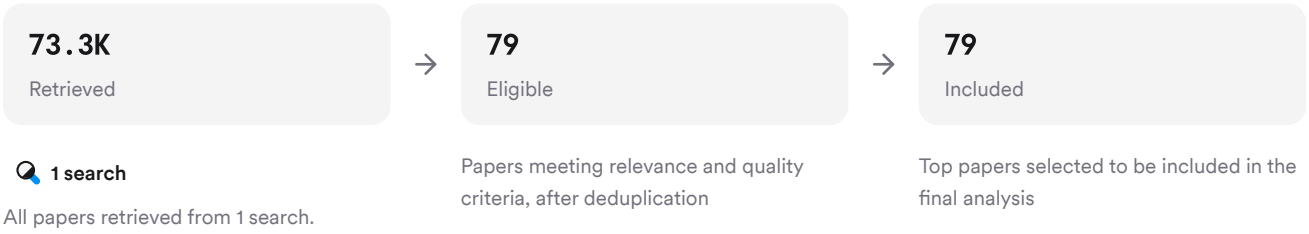


FIGURE 2 Deep Search paper selection funnel

Searches targeted metric definitions, formulas, retrieval scoring, forgetting, temporal consistency, conflict handling, and efficiency reporting for long-horizon agents.

3. Results

3.1 Key Papers

A few papers anchor the metric landscape because they either catalog the main metric families or define reusable benchmark suites. The TMLR survey and the 2026 survey by Du synthesize the standard stack; MemBench, MemoryAgentBench, AgentMemBench, MemGUI-Bench, MemConflict, and SeqMem-Eval contribute many of the most explicit metric definitions (Huang et al., 2026; Du, 2026; Tan et al., 2025; Liu et al., 2026; Cherif, 2026; Tao et al., 2026; Dong et al., 2026).

Paper	Summary
(Huang et al., 2026)	Survey of common memory metrics (Huang et al., 2026)
(Cherif, 2026)	Retrieval, answer, efficiency formulas (Cherif, 2026)
(Tao et al., 2026)	Conflict-aware memory diagnostics (Tao et al., 2026)

FIGURE 3 Foundational papers for memory metric definitions

3.2 Core Formulas

The most standardized formulas are inherited from information retrieval and QA. **Accuracy / Memory Accuracy** is the proportion of correctly answered instances, typically $ \mathrm{Accuracy}=\frac{1}{N}\sum_{i=1}^{N}\mathbf{1}[\hat a_i=a_i] $ with (N) questions, ground truth (a_i), prediction (\hat a_i), and indicator (\mathbf{1}) (Huang et al., 2026; Zhang et al., 2024; Tan et al., 2025). **Precision**, **Recall**, and **F1** for retrieved evidence or token overlap use the standard formulas $ P=\frac{TP}{TP+FP},\quad R=\frac{TP}{TP+FN},\quad F1=\frac{2PR}{P+R} $ and surveys explicitly note token-level F1 as the harmonic mean of precision and recall (Huang et al., 2026; Zhang et al., 2024; Cherif, 2026; Yu et al., 2026).

Recall@K is usually $ \mathrm{Recall@}K=\frac{|Retrieved_{1:K}\cap Relevant|}{|Relevant|} $ or, when only one gold turn exists, an indicator that the reference appears in the top (K) retrieved memories (Huang et al., 2026; Cherif, 2026; Liu et al., 2026). **MRR** is the mean reciprocal rank of the first relevant result, $ \mathrm{MRR}=\frac{1}{|Q|}\sum_{q}\frac{1}{rank_q}, $ and **MAP** averages precision at relevant ranks across queries (Huang et al., 2026; Cherif, 2026; Li et al., 2026). **nDCG@K** is normalized discounted cumulative gain over the top (K), typically $ \mathrm{nDCG@}K=\frac{DCG@K}{IDCG@K},\quad DCG@K=\sum_{i=1}^{K}\frac{2^{rel_i}-1}{\log_2(i+1)}, $ used when relevance is graded rather than binary (Cherif, 2026; Huang et al., 2026).

Some benchmarks adapt these base formulas to tool use or interactive success. **Tool Accuracy** is 1 only when the correct tool is invoked and all parameters match exactly, while parameter-level F1 compares predicted and gold tool arguments (Shen et al., 2026). **Pass@1** and **Pass@K Success Rate** measure whether an agent succeeds on the first attempt or within (K) attempts across repeated trials (Liu et al., 2026).

3.3 Specialized Memory Metrics

Several recent papers introduce metrics that target memory failures missed by recall and F1. Memora introduces **Forgetting-Aware Memory Accuracy (FAMA)**, which starts from answer correctness but penalizes use of obsolete or invalidated memory, showing that standard memory-presence evaluation overestimates long-term performance (Uddin et al., 2026). MemConflict adds **Conflict Recognition Score (CRS)**, **Update Order Consistency Score (UOCS)**, **Support Evidence Hit@K (SEH@K)**, and **Support Rank Score (SRS)** to separate final answer quality from whether the valid memory was retrieved and ranked above stale or conditionally invalid alternatives (Tao et al., 2026).

MemGUI-Bench defines **Information Retention Rate (IRR)** as the proportion of required information units correctly recalled and used, **Memory-Task Proficiency Ratio (MTPR)** as performance on memory-intensive tasks relative to standard tasks, and **Failure Recovery Rate (FRR)** as a harmonic-decay-weighted measure rewarding faster recovery from failure across attempts (Liu et al., 2026). StreamMemBench adds four stagewise diagnostics: evidence recall, initial evidence use, feedback incorporation, and follow-up reuse, explicitly measuring whether stored evidence improves future-oriented assistance (Liu et al., 2026). MemGround introduces **Memory Fragments Unlocked (MFU)** and **Memory Fragments with Correct Order (MFCO)** to quantify how much of the required memory has been surfaced and whether temporal sequence was preserved (Ding et al., 2026).

SeqMem-Eval shifts evaluation toward continual-learning style dynamics by measuring **online utility**, **hold-out generalization**, **backward transfer**, and **forgetting** over evolving memory states rather than just final accuracy (Dong et al., 2026). SeqMem-Eval's key contribution is not a new retrieval formula but a longitudinal view: high final performance can coexist with strong forgetting or negative transfer (Dong et al., 2026).

3.4 Efficiency, Governance, and Safety Metrics

A distinct metric family measures whether memory is practical to deploy. **Memory Footprint** is usually the average number of memory-context input tokens per evaluation turn, and **Per-turn Latency** is end-to-end wall-clock time per turn (Cherif, 2026). MemBench adds **Memory Capacity**, defined operationally as the threshold where performance drops sharply as memory content grows, plus **temporal efficiency** in read/write processing (Tan et al., 2025). MemGUI-Bench further reports **Average Step Ratio**, **Average Time per Step**, and **Average Cost per Step** (Liu et al., 2026).

Several papers argue that write-path and management costs should be first-class metrics. MemDelta recommends reporting write-path cost, because embedding choice or construction overhead can flip conclusions without changing headline accuracy (Wang, 2026). Harness-based evaluations also call for measuring LLM calls, ingestion cost, management overhead, and compression ratio, since many published systems omit them (Huang et al., 2026).

Governance and privacy metrics are newer but increasingly explicit. Deployment-time memorization work defines **Personalization Recall (PR)**, **Adversarial Extraction Rate (AER)**, **Privacy-Utility AUC**, and **Forgetting Residue Score (FRS)** to quantify the privacy-utility frontier and post-deletion leakage from derived memory tiers (Chen et al., 2026). FiFA evaluates **narrative coherence**, **goal completion rate**, **social recall accuracy**, **privacy leakage rate per turn**, and **cost efficiency** under token budgets (Alqithami, 2025). Memory governance papers also propose per-memory value signals such as **Memory Worth**, which converges to [p^+(m)=\Pr[y_t=+1\mid m\in M_t],] the empirical conditional success probability given that memory (m) was retrieved (Şimşek, 2026).

Results Timeline

Agent memory metrics have broadened from recall to lifecycle evaluation.

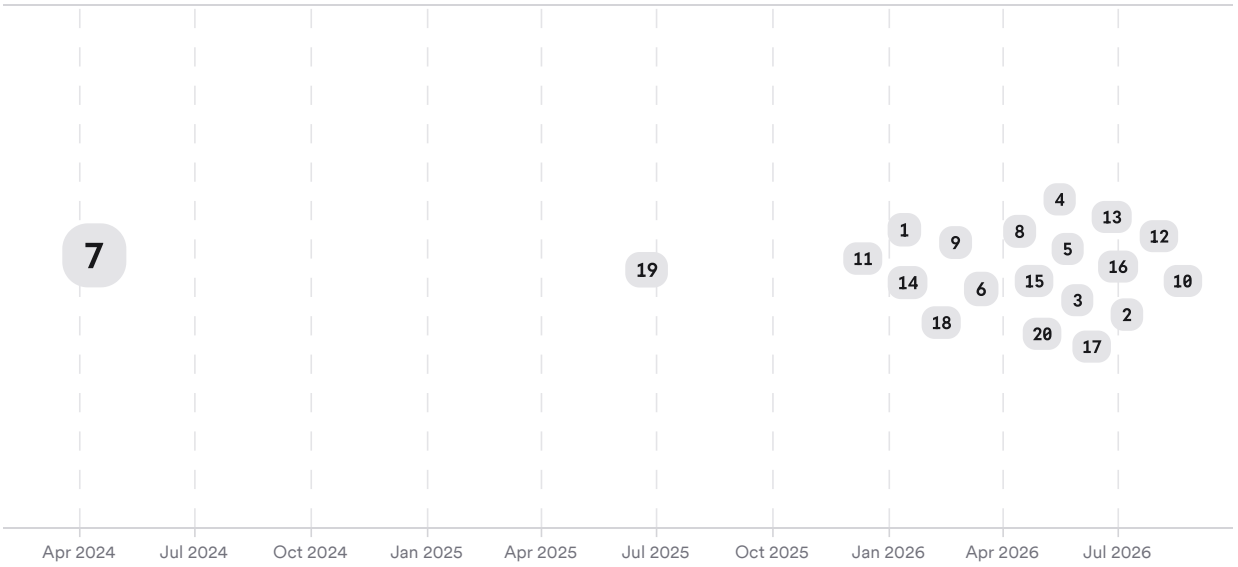


FIGURE 4 Timeline of agent memory benchmark evolution; larger markers indicate more citations

The timeline shows a clear arc: early benchmarks emphasized recall and QA, 2025 added capacity and interactive multi-turn evaluation, and 2026 brought conflict-aware, privacy-aware, multimodal, artifact-auditing, and scale-conditioned metrics (Wu et al., 2024; Tan et al., 2025; Hu et al., 2025; Tao et al., 2026; Guo et al., 2026; Shao et al., 2026).

Top Contributors

Based on this corpus, **Yu Wang**, **Wei-Chieh Huang**, and **Julian McAuley** appear most often in benchmark and survey papers on agent memory metrics, ranked by repeated appearance within these included papers rather than field-wide impact (Hu et al., 2025; Huang et al., 2026; Enze et al., 2026).

Type	Name	Papers
Author	Yu Wang	[4d575f6793445919a286bf2779248da6] [701ab9bb15c15699bd5b339e0703a107] [c68a330378225b2d93b89832da886fc1] [2bdd2af41da05d2eb8aafa5d78e8b867] [218a00e4d9ee58ae9b3a147ddbdf489e]
Author	Wei-Chieh Huang	[701ab9bb15c15699bd5b339e0703a107] [b1fb4a2a94e75636b980d57f8fab25ec] [65220c42d93c56eeada1cb366ab1bafc]
Author	Julian McAuley	[4d575f6793445919a286bf2779248da6] [701ab9bb15c15699bd5b339e0703a107] [c68a330378225b2d93b89832da886fc1]

Type	Name	Papers
Journal	*ArXiv*	[4d575f6793445919a286bf2779248da6][9bece15d5ef95791a76b3621f17a6396] [fcb6253470895805b58deaa3daa11ff9] [c68a330378225b2d93b89832da886fc1] [2bdd2af41da05d2eb8aafa5d78e8b867]
Journal	*Trans. Mach. Learn. Res.*	[701ab9bb15c15699bd5b339e0703a107]
Journal	*ACM Transactions on Information Systems*	[49b68544092450f9a9c74be746426a4f]

FIGURE 5 Authors and journals appearing most often

4. Discussion

The evidence strongly supports a **metric stack** rather than a single benchmark score. Classical IR formulas remain the backbone for memory retrieval evaluation, and their definitions are stable across surveys and benchmarks (Huang et al., 2026; Zhang et al., 2024; Cherif, 2026). But those metrics only answer whether relevant evidence was surfaced, not whether it was current, ranked above conflicts, safely forgotten, or used correctly in action (Wu et al., 2026; Long et al., 2026; Tao et al., 2026).

The next major shift is from **answer-level** to **memory-level** and **trajectory-level** evaluation. Fact-level probing, stagewise diagnostics, and recovery-style auditing all try to localize failure to storage, updating, retrieval, ranking, or use (Long et al., 2026; Liu et al., 2026; Enze et al., 2026; Du et al., 2026). This is important because several papers show that evidence is often retrievable but still misused, and because benchmark rankings can reverse when history length, interference, or operating budget changes (Long et al., 2026; Shao et al., 2026; Spencer, 2026).

A limitation is that many newer metrics are benchmark-specific and lack universally adopted formulas. FAMA, IRR, MTPR, FRR, MFU, MFCO, CRS, UOCS, PR, AER, FRS, and Memory Worth each target a real failure mode, but cross-paper comparability remains limited because implementations and aggregation schemes differ (Uddin et al., 2026; Liu et al., 2026; Ding et al., 2026; Tao et al., 2026; Chen et al., 2026; Şimşek, 2026).

Claim	Evidence Strength	Reasoning	Papers
Memory evaluation needs multiple metrics beyond accuracy	Strong	Repeated across surveys, benchmarks, and critiques	(Huang et al., 2026; Zhou et al., 2026; Alqithami, 2025)
Retrieval formulas are standardized and widely shared	Strong	Accuracy, F1, Recall@K, MRR, MAP, nDCG recur with stable definitions	(Huang et al., 2026; Zhang et al., 2024; Cherif, 2026)

Claim	Evidence Strength	Reasoning	Papers
Evidence-use and conflict metrics fill major blind spots	Moderate	Several 2026 benchmarks add explicit retrieval-versus-use diagnostics	(Tao et al., 2026; Liu et al., 2026; Long et al., 2026)
Efficiency and write-path cost are underreported but essential	Strong	Many systems omit latency, footprint, or construction cost	(Zhou et al., 2026; Huang et al., 2026; Wang, 2026)
Governance metrics are promising but not yet standardized	Moderate	Privacy, deletion, and per-memory value metrics are newer and less replicated	(Chen et al., 2026; Şimşek, 2026; Rusu et al., 2026)

FIGURE 6 Claims and evidence across memory metric literature

5. Conclusion

Agent memory evaluation now spans a broad catalog of metrics with partially standardized formulas. The mature core is still **Accuracy, Precision, Recall, F1, Recall@K, MRR, MAP, and nDCG**, but current research increasingly adds forgetting-aware, conflict-aware, use-aware, privacy-aware, and efficiency-aware measures because retrieval accuracy alone misses major failure modes.

Research Gaps

The coverage is broad on text retrieval and conversational QA, but much thinner on standardized update-governance and evidence-use metrics across environments.

	Answer Metrics	Ranking Metrics	Temporal Validity	Efficiency Cost	Formal Formula
Retrieval Fidelity	GAP	GAP	GAP	GAP	GAP
Selective Forgetting	GAP	GAP	GAP	GAP	GAP
Conflict Handling	GAP	GAP	GAP	GAP	GAP
Action Grounding	GAP	GAP	GAP	GAP	GAP

	Answer Metrics	Ranking Metrics	Temporal Validity	Efficiency Cost	Formal Formula
Privacy Governance	GAP	GAP	GAP	GAP	GAP

FIGURE 7 Coverage gaps across memory metric families

The biggest gap is **standardization of non-retrieval metrics**. Retrieval metrics have portable formulas used across many papers, but update-validity, forgetting, privacy, and action-grounding metrics are fragmented into benchmark-local definitions (Cherif, 2026; Uddin et al., 2026; Tao et al., 2026; Shen et al., 2026; Chen et al., 2026). Another gap is that several papers identify **evidence use** as the dominant bottleneck, yet few offer broadly reused formulas for it (Long et al., 2026; Liu et al., 2026; Tao et al., 2026). Multimodal, embodied, and domain-specific settings are also adding bespoke metrics faster than the field is harmonizing them (Guo et al., 2026; Rasheed et al., 2026; Du et al., 2026).

Open Research Questions

Future work is moving toward reusable lifecycle metrics that transfer across conversational, tool-based, multimodal, and embodied agents.

Question	Why
How should evidence use after retrieval be measured with a standardized formula?	Many systems fail despite retrievable evidence, but usage metrics remain benchmark-specific.
Which forgetting and update metrics best capture stale-memory failures across domains?	FAMA, conflict scores, and temporal-validity measures exist, but they are not yet unified.
How should memory evaluation combine quality, privacy, and operational cost in one comparable protocol?	Deployment papers now report leakage, residue, latency, and footprint, but no common reporting standard exists.

The most important shift is from **memory-as-recall** to **memory-as-lifecycle evaluation**; the biggest open question is how to standardize evidence-use and forgetting metrics across agent settings.

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at https://consensus.app. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

References

E., Zhou, Y., Huang, W.-C., Yang, J., H., Wang, Z.-X., Li, C., Miao, C., Yu, P. S., & Wang, Z. (2026). MEMPROBE: Probing Long-Term Agent Memory via Hidden User-State Recovery. *ArXiv, abs/2606.24595*. https://doi.org/10.48550/arxiv.2606.24595

Alake, R., Bernardis, C., Cayet, P., Engel, L., Hilloulin, D., Hong, S., Hosler, A., Kavantzas, N., Kossyk, I., Le, S., Patra, R., Talamadupula, K., & Venzin, V. (2026). Oracle Agent Memory as an Enterprise Memory Substrate for Long-Horizon AI Agents. *ArXiv, abs/2607.13157*. https://doi.org/10.48550/arxiv.2607.13157

Alqithami, S. (2025). Forgetful but Faithful: A Cognitive Memory Architecture and Benchmark for Privacy-Aware Generative Agents. *ArXiv, abs/2512.12856*. https://doi.org/10.48550/arxiv.2512.12856

Chen, L., Zhang, G., Zhao, K., Cirne, D., Olsen, A., Chu, X., Miller, Z., Blanken, A., Anoun, A., & Ting, J. (2026). Deployment-Time Memorization in Foundation-Model Agents. *ArXiv, abs/2606.10062*. https://doi.org/10.48550/arxiv.2606.10062

Cherif, A. (2026). AgentMemBench: A Systematic Benchmark for Evaluating Long-Term Memory Management Strategies in Conversational AI Agents. https://doi.org/10.48550/arxiv.2608.00009

Ding, Y., Xia, W., Zhao, Y.-T., Su, J., Yang, J., Zhang, Z., & Yang, W. (2026). MemGround: Long-Term Memory Evaluation Kit for Large Language Models in Gamified Scenarios. *ArXiv, abs/2604.14158*. https://doi.org/10.48550/arxiv.2604.14158

Dong, S., Chen, Z., Shi, C., Wang, P., Li, J., & Shen, C. (2026). Is One Score Enough? Rethinking the Evaluation of Sequentially Evolving LLM Memory. *ArXiv, abs/2605.15384*. https://doi.org/10.48550/arxiv.2605.15384

Du, P. (2026). Memory for Autonomous LLM Agents:Mechanisms, Evaluation, and Emerging Frontiers. *ArXiv, abs/2603.07670*. https://doi.org/10.48550/arxiv.2603.07670

Du, C., Vosseler, A., Mazza, F., & Borrmann, A. (2026). IFCMemoryBench: Evaluating Long-Term Memory of LLM-Based Agents in BIM Information Retrieval. *ArXiv, abs/2607.26072*. https://doi.org/10.48550/arxiv.2607.26072

Guo, M.-H., Jiao, Q., Shi, Z., Quan, Y., Zhang, B., Li, D., Che, L., Xu, W., Liu, S., Liu, Z., Kapadia, M., Pavlovic, V., Liu, J., Wang, M., Shi, Y., Metaxas, D. N., & Tang, R. (2026). MemEye: A Visual-Centric Evaluation Framework for Multimodal Agent Memory. *ArXiv, abs/2605.15128*. https://doi.org/10.48550/arxiv.2605.15128

Hu, Y., Wang, Y., & McAuley, J. (2025). Evaluating Memory in LLM Agents via Incremental Multi-Turn Interactions. *ArXiv, abs/2507.05257*. https://doi.org/10.48550/arxiv.2507.05257

Huang, W.-C., Zhang, W.-Z., Liang, Y., Bei, Y., Chen, Y., Feng, T., Pan, X.-Y., Tan, Z., Wang, Y., Wei, T., Wu, S., Xu, R., Yang, L., Yang, R., Yang, W., Yeh, C.-Y., Zhang, H.-R., Zhang, H., Zhu, S.-Q., . . . Shu, K. (2026). A Survey of Agent Memory in the Second Half: Towards Self-Evolving and Long-Horizon Agents. *Trans. Mach. Learn. Res., 2026*. https://doi.org/10.48550/arxiv.2602.06052

Huang, W.-C., Zhang, W., Wu, Y.-C., Chen, Y., Jiang, E., Yang, W., Yang, Y., Zou, H. P., Zhang, H.-R., Wu, Y., Wu, H., Chang, K.-W., Yu, P. S., Liu, X., & Caliskan, A. (2026). Harness the Memory: A Holistic Evaluation of Memory Substrates in Memory Agents. https://doi.org/10.48550/arxiv.2608.15008

Jiang, D., Li, Y., Wei, S.-T., Yang, J., Kishore, A., Zhao, A., Kang, D., Hu, X., Chen, F., Li, Q., & Li, B. (2026). Anatomy of Agentic Memory: Taxonomy and Empirical Analysis of Evaluation and System Limitations. *ArXiv, abs/2602.19320*. https://doi.org/10.48550/arxiv.2602.19320

Li, C., Zhang, M., Kang, J., Chen, D., Shen, J., Tang, B., Zhou, X., Xiong, F., & Li, Z. (2026). MemReranker: Reasoning-Aware Reranking for Agent Memory Retrieval. *ArXiv, abs/2605.06132*. https://doi.org/10.48550/arxiv.2605.06132

Liu, C., Yang, Y., Pu, S. X., Liu, Y., Long, L., Guo, Y., Chen, N., Weng, Z.-T., Kochkina, E., Kaur, S., Smiley, C. H., Liu, X., Zou, J., Liu, S., Bu, Y., Peng, S., & Wang, X. (2026). WorldMemArena: Evaluating Multimodal Agent Memory Through Action-World Interaction. *ArXiv, abs/2605.29341*. https://doi.org/10.48550/arxiv.2605.29341

Liu, G., Ren, Y., Gu, H., Zhang, P., Wang, W., Liu, J.-H., Gu, N., & Lu, T. (2026). StreamMemBench: Streaming Evaluation of Agent Memory for Future-Oriented Assistance. *ArXiv, abs/2606.14571*. https://doi.org/10.48550/arxiv.2606.14571

Liu, G.-Y., Zhao, P., Liang, Y., Luo, Q., Tang, S., Chai, Y., Lin, W., Xiao, H., Wang, W., Chen, S., Lu, Z., Wu, G., Wang, H., Liu, L., & Liu, Y. (2026). MemGUI-Bench: Benchmarking Memory of Mobile GUI Agents in Dynamic Environments. *ArXiv, abs/2602.06075*. https://doi.org/10.48550/arxiv.2602.06075

Liu, S., Zhu, J., Shu, L., Lin, J., Chen, Y., Zhang, H., Zhang, C., Xu, D., Li, J., Tang, B., Li, Z., Xiong, F., Chen, E., & Xu, T. (2026). PERMA: Benchmarking Personalized Memory Agents via Event-Driven Preference and Realistic Task Environments. *ArXiv, abs/2603.23231*. https://doi.org/10.48550/arxiv.2603.23231

Long, X., Chen, Z., Zeng, S., Wang, S., Guo, K., & Tang, J. (2026). MemTrace: Probing What Final Accuracy Misses in Long-Term Memory. *ArXiv, abs/2606.17328*. https://doi.org/10.48550/arxiv.2606.17328

Rasheed, A., Automatika, M. K., & Inria, R. (2026). eMEM: A Hybrid Spatio-Temporal Memory System For Embodied Agents. *ArXiv, abs/2606.03374*. https://doi.org/10.48550/arxiv.2606.03374

Rusu, T., Khanzadeh, S., & Alalfi, M. H. (2026). Selective Forgetting: A Graph-Based Memory Framework for Long-Term LLM Agents. https://doi.org/10.48550/arxiv.2608.28978

Shao, J.-Q., Lu, Y., Zhang, Y., & Luo, B. (2026). When Stored Evidence Stops Being Usable: Scale-Conditioned Evaluation of Agent Memory. *ArXiv, abs/2605.07313*. https://doi.org/10.48550/arxiv.2605.07313

Shen, Y., Li, K., Zhou, W., & Hu, S. (2026). Mem2ActBench: A Benchmark for Evaluating Long-Term Memory Utilization in Task-Oriented Autonomous Agents. *ArXiv, abs/2601.19935*. https://doi.org/10.48550/arxiv.2601.19935

Şimşek, B. (2026). When to Forget: A Memory Governance Primitive. *ArXiv, abs/2604.12007*. https://doi.org/10.48550/arxiv.2604.12007

Spencer, Q. H. (2026). Ground Truth First: A Longitudinal Evaluation Instrument for Agent Memory, and the Tenure Crossover in Memory-Architecture Rankings. *ArXiv, abs/2607.21962*. https://doi.org/10.48550/arxiv.2607.21962

Tan, H.-R., Zhang, Z., C., Chen, X., Dai, Q., & Dong, Z. (2025). MemBench: Towards More Comprehensive Evaluation on the Memory of LLM-based Agents. 19336-19352. https://doi.org/10.48550/arxiv.2506.21605

Tao, Z., Zhao, J., Liu, P., Xi, D., Chen, Y., Xu, W., & Li, Z. (2026). MemConflict: Evaluating Long-Term Memory Systems Under Memory Conflicts. *ArXiv, abs/2605.20926*. https://doi.org/10.48550/arxiv.2605.20926

Uddin, M. N., Shubham, K., Blanco, E., Baral, C., & Wang, G. (2026). From Recall to Forgetting: Benchmarking Long-Term Memory for Personalized Agents. *ArXiv, abs/2604.20006*. https://doi.org/10.48550/arxiv.2604.20006

Wang, K. (2026). MemDelta: Controlled Baselines and Hidden Confounds in Agent Memory Evaluation. *ArXiv, abs/2606.29914*. https://doi.org/10.48550/arxiv.2606.29914

Wu, D., Wang, H., Yu, W., Zhang, Y., Chang, K.-W., & Yu, D. (2024). LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory. *ArXiv, abs/2410.10813*. https://doi.org/10.48550/arxiv.2410.10813

Wu, D., Ji, Z.-X., Kawatkar, A., Kwan, B., Gu, J.-C., Peng, N., & Chang, K.-W. (2026). LongMemEval-V2: Evaluating Long-Term Agent Memory Toward Experienced Colleagues. *ArXiv, abs/2605.12493*. https://doi.org/10.48550/arxiv.2605.12493

Yu, J.-W., Fang, Y., Liu, X., & Y. (2026). H-Mem: A Novel Memory Mechanism for Evolving and Retrieving Agent Memory via a Hybrid Structure. *ArXiv, abs/2605.15701*. https://doi.org/10.48550/arxiv.2605.15701

Zhang, Z., Dai, Q., Bo, X., C., Li, R., Chen, X., Zhu, J., & Dong, Z. (2024). A Survey on the Memory Mechanism of Large Language Model-based Agents. *ACM Transactions on Information Systems, 43*, 1 - 47. https://doi.org/10.1145/3748302

Zhou, W., Zhou, X.-H., Han, S.-K., Xu, H.-M., Li, G., Li, Z., Xiong, F., & Wu, F. (2026). Are We Ready For An Agent-Native Memory System?. *ArXiv, abs/2606.24775*. https://doi.org/10.48550/arxiv.2606.24775